

Tony Siu

FOUNDING AI ENGINEER · MULTI MODAL GENAI RESEARCHER · APPLIED STATISTICS RESEARCHER

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Engineering Experience

Founding AI Engineer & Researcher

San Francisco, CA

A16z Speedrun SR005

Apr 2025 - Dec 2025

- Principal Engineer – 60% singlehanded engineering output in a company of 20 people
- 200x optimization – 0.2 sec per 78+ VS 5 sec per 1 animation rendering for spritesheets, .gifs
- Architect & built – Natural language to full GenAI game generation MVP in under a month
- Develop Vision based Promptless UI action Agentic systems for long-context inter-session user profiles.
- Reimplement Bleeding Edge papers such as the Agentic Context Engineering(ACE) paper or Tongyi labs AgentFold paper as CUAs Scaffolds

Generative AI Engineer

Philadelphia, PA

A Pro Inc Solutions

Feb 2025 - July 2025

- Architect end to end systems to generate AI Influencers, Key Opinion Leaders for AI Digital Marketing as a service.
- Design application evaluation criterias to determine the appropriateness of the Generated Videos, Audio, Images of the AI Influencer.
- 100x cost savings & Orchestration of distributed training and inference pipelines
- Cost free Preemptible AI infrastructure hosting with savvy tooling and UIUX design
- Research & Engineer upon SOTA Multi-Modal, Multi Step Generative models with preference optimization to align generated AI influencers

Financial Data Engineer

Philadelphia, PA

Temple Fox School of Business

Jul 2023 - Jan 2025

- Increased 200,000 SOI data samples from 20 different BDCs to model SEC registered BDCs fiscal product categorization & valuation
- Eliminated need to post process data using OpenAIs gpt4o-mini, reducing data pipeline staging processes by 50%
- Prototyped and implemented staged Data models for ELT Data pipeline and Database for single source of truth mapping
- Containerized data extraction and data cleaning programs for future use and cross platform portability

Data Analytics Intern

Philadelphia, PA

Symetri USA

Jun 2024 - Sept 2024

- Abstracted away traditional data infrastructure using AlaaS paradigm, maintaining client data security and minimal technological overhead
- Migrated data infrastructure to Multi-Agent GenAI BIM Vector and Graph based Retrieval Augmented Generation POCs for client demonstration
- Landed partnerships with PennMedicine, several other Hospital Authorities and Port Authorities in PANJNY through built Compliance Frontend

Analytics Engineer Intern

July 2021 - Sept 2021

Fio.one

Taipei,Taiwan

- End-to-end Recommendation System integrating layer 2 blockchain protocols with layer 3 Web UI interfaces, Blockchain with Data Science
- Design ERM and Database administration policies for highschool students tagged to a recommended university's departments.
- Decoded combinations of big-5, utf-8, and Cp1252 data encodings to unify scraped data from different languages

AI Specialist Engineer

Taipei, Taiwan

Taiwan Tech Arena

May 2020 - Oct 2020

- Reduced fps stitching speed of 2x 3840X2160 pixels, 170 degree fisheye cameras from 6 seconds to 0.2 seconds on the Nvidia Jetson Xavier
- Reduced 0.2 to 0.01 fps using CUDA code integration of opencv 2.4 image blending into OpenCV 4.5 stitching during composePanorama
- EstimateTransform only once for real time composePanorama with multi-threading as camera is stationary but capture is 360 degrees
- 6x reduction in opencv cuda/cudnn binary compilation time by generating CPack binary deb files for single Opencv compilation (6 to 1 hours)
- Real time 3d video to 2d image projection homographies of player field positions
- Real time automated fisheye to panoramic view camera callibration to reconstruct 360 camera view to landscape view of sports field
- 6 main analysis features of mainly "camera panning", "zooming" and "slow motion"
- Ported Opencv C++ CUDA code between aarch64 tegra processors and x64-86 processors
- Patched opencv C++ customly optimized source code to opencv-python API via .so source file symlinks
- Containerized optimized Opencv custom source code for cross platform operations
- Automated video delivery and running of analysis from edge device to cloud instance via Apache Airflow dags

Education

Masters, Computational Data Science | GPA 4.0

Philadelphia, PA

Temple University

Jan 2024 - On Hold

- IEEE Cloud & AI Student Ambassador
- Accelerated 4+1 Masters Computational Data Science program
- Promoted establishment of Google Developers Student Clubs at Temple

Bachelors, Computational Data Science | GPA 3.87

Philadelphia, PA

Temple University

Aug 2022 - Dec 2024

- Graduated with Distinction
- Dean's list Fall 2024
- Visual Question Answering Preference Optimization Research under Dr Longin Jan Latecki

Research Experience

Multi-Modal Generative AI Researcher

Philadelphia, PA

Temple College of Science & Technology

May 2023 - May 2025

- Multiple papers to IEEE, CAIP and ECAI, prestigious Computer Vision and AI research conferences
- 6% performance increase via rPPG extracted from RhythmFormer & RhythmMamba for Engagement Video QA to baseline finetuning
- 6% performance increase using Direct preference optimization to generalize to both in-sample, out-of-sample data to baseline finetuning
- 9% performance increase with latest Preference Alignment algorithms to baseline i.e. KTO,IPO,RSO,cDPO,SSPO,GRPO
- Self generated preferred/rejected response pairs without human annotation for Direct preference Optimization without RLHF
- Performed experiments on DAiSEE, EngageNet, SED Student Engagement datasets with MiniGPT-4-Video, Llava-Next, VideoMamba
- Achieved 97% F1 precision recall on student engagement dataset using Engagement Level prompt finetuning for Visual QA task

Google UR Computer Vision Research Lead

Taipei, Taiwan

Google Research

Jan 2022 - May 2023

- Led Computer Vision research at National Taiwan University to enhance Googles testing suite for detecting UI defects on their Pixel Mobile
- Secured further Google level 2 sponsorship at National Taiwan University through pioneering research in to Multi-Modal Computer Vision
- 43% benchmark to 98% precision-recall enhancement for UI Flicker Detection using CNN-Transformers model
- 10x increase in data augmented samples, 180 to 1991 videos snippets
- 20x reduction in model training time through appropriate initialization of weights and bias, 1000 epochs to just 50 epochs

Research Associate

Taipei, Taiwan

Smart Contract Research Forum

Oct 2021 - Apr 2022

- Bounties on "Do you need a blockchain?", "Towards private on-chain algorithmic trading", "Machine Learning Guided Cross-Contract Fuzzing"
- Published Research Articles, discussion posts, research pulses, podcasts, and community calls as SCRF moderator for APAC region
- Acted as the Moderator for SRCF engagement for the APAC region

Publications

Pang, L., **Siu, T.**, Alazzawe, A., Kant, K., Latecki, L.J. (2026). Generalizable Detection of Student Engagement in On-line Learning Environments. In: Castrillón-Santana, M., et al. Computer Analysis of Images and Patterns. CAIP 2025. Lecture Notes in Computer Science, vol 15622. Springer, Cham. 

P.P Kumar, **Siu. T.**, S. Millwood, M.M Rogalska, K. Krishna and L.J. Latecki (2024). "Exploiting rPPG Signal Enhances Student Engagement Recognition in Online Video Education" WACV, 2025. 

Community Outreach

Code&Coffee Philadelphia Chapter Founder

Philadelphia, PA

Code&Coffee

Jan 2024 - Present

- Founded the only consistent builders tech focused meetup in Philadelphia as a non profit organization
- Grew from 2 founding members to 3100+ members in a little more than a year
- Connect Engineers, Founders, and Researchers from many Technology focused domains
- Host events ranging from AI paper discussions to Tech Founders coworking to LeetCode practice sessions to casual coffee and code events
- Host the only builders Hackathon that took place during Philly Tech Week 2025

Google Community Lead

Taipei, Taiwan

Google Developers Groups

Jan 2020 - Dec 2023

- Google AI Explore ML with Crowd Source Facilitator
- Google I/O Taiwan 2023 Speaker
- Google Devfest '21 Student Speaker & Panelist & Event Volunteer
- Google Developers Student Club Lead Founder, TamKang University
- Google Developers Student Club Core Member, National Taiwan Normal University

Skills

Work Focus	Multi-modal Generative AI, Computer Vision, Natural Language processing, Statistical Analysis, Data Engineering, Retrieval-Augmented Generation (RAG), Cloud System Administration
Programming	C, C++, CUDA, Rust, Java, JavaScript, TypeScript, Python, R, SQL, Bash ,Latex
Machine Learning	Pytorch, Tensorflow, Keras, Opencv, NLTK, Huggingface, Scikitlearn, OpenAI, Xgboost, Statsmodels, LangChain, LlamaIndex, Haystack, Hyperopt, Optuna
Scientific Computing	Pandas, Numpy, Scipy, Sympy
Data Visualization	Dash Plotly, Matplotlib, Streamlit, Paraview, VTK, D3.js, PowerBI
Infrastructure	Linux, Git, AWS, GCP, Docker, MySQL, MongoDB, PostgreSQL, Neo4j, ChromaDB, Apache Airflow, PowerAutomate, Docker, Docker-compose
WebDev	FastAPI, Django, Flask, Express.js, D3.js